FORLAIS GROUP | ACADEMIC PAPER

BF16 1.52x Entropy Ceiling

Evidence-grounded FCN academic paper

Oliver Cooper

Forlais Group Ltd, London, United Kingdom

ocooper@forlaisgroup.com

ORCID: 0009-0005-5798-827X

PAPER REFERENCE	**FCN-C22-X152-V-ENTROPY-CEILING**
VERSION	1.0-public
DATE	27 June 2026
STATUS	Public Release
CLASSIFICATION	Public
CORRESPONDENCE	Oliver Cooper, Forlais Group Ltd
REVIEW OWNER	Research Disclosure Review
DISCLOSURE STATUS	Approved for public release
RELEASE CHANNEL	GitHub + Zenodo
PUBLICATION TARGET	GitHub + Zenodo public archive

Abstract

We report verified evidence for BF16 1.52x Entropy Ceiling within the Forlais Compression Notation (FCN) research programme. The result has an associated compression ceiling of approximately $1.522\times$ in the tested setting, but the principal contribution of this paper is the empirical regularity itself rather than any single codec ratio. The associated reference entropy is approximately 10.51 bits per value. Measurements are taken in the BF16 setting. Numerical values are reported at the abstraction level approved for public disclosure; implementation details, run procedures, and operational artefacts are out of scope here. The claim is scoped to the tested setting and does not extend to lossy compression, learned non-IID coders, format redesign, or task-specific model compression unless explicitly stated.

Keywords: lossless compression; neural network weights; Forlais compression research; BF16; Huffman coding; arithmetic coding

Methodology Disclosure Boundary

This paper states the FCN evidence for the reported result at a public-safe abstraction level. It reports measured quantities, reported scope, and limitations. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

Lossless compression of trained neural network weights is studied across the FCN research programme as a coding question first and an engineering question second. For each FCN entry, the goal is to state the smallest public-safe claim that the evidence actually supports, with the format, scope, and reconstruction target made explicit.

This paper reports the FCN evidence for BF16 1.52x Entropy Ceiling. The result is treated as a fundamental property of the trained-weight distribution, not as a property of any one codec or implementation.

The headline numbers, all sourced from the FCN evidence register, are a reported ratio of approximately $1.522\times$, a measured entropy of approximately 10.51 bits per value.

The remainder of the paper states the public-safe scope, summarises the relevant background, presents the measurement-level evidence, separates verified facts from speculative interpretation, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- BF16 stored weights;
- value-level entropy measurement of BF16 stored weights;
- public-safe summarisation of measured quantities.

The result does not concern:

- lossy model compression, pruning, distillation, and quantisation-aware retraining as separate regimes;
- learned non-IID coders that operate over broader joint distributions than the tested setting;
- format redesign or architecture-specific compression that requires modifying model semantics;
- cross-model delta regimes, which are reported in separate FCN entries;
- content-addressable deduplication, which is reported in separate FCN entries.

Each excluded regime is a distinct setting with its own evidence base. Ratios from those settings should not be compared with the headline number in this paper without first reconciling assumptions about coding scope, format, and reconstruction target.

3 Background and related work

Lossless compression of trained neural network weights sits at the intersection of classical source coding and contemporary neural network engineering. Shannon's source coding theorem gives a lower bound on the expected code length of any lossless code in terms of the entropy of the source distribution [12, 2].

Huffman coding and arithmetic coding provide practical means to approach the entropy bound on a discrete source [3, 13, 11, 10].

Modern model weights are typically stored in BF16 or FP32. The relevant numeric formats are specified in the IEEE 754 standard and discussed in the BF16 and mixed-precision training literature [4, 6, 8].

Algebraic structure in finite-field representations is captured classically by linear-recurrence and shift-register synthesis [1, 7].

We refer to the publicly released model artefacts associated with the architectures examined in this evidence set [5, 9].

4 Evidence summary

The numerical summary in this section is reproduced from the internal FCN evidence register. Values appear here only at the public abstraction level. The release excludes underlying implementation details, run procedures, and source logs.

Reported quantity	Public-safe value
Reported compression ratio	approximately 1.522x
Total entropy	approximately 10.51 bits per value
Internal measurement date	2026-02-14

Table 1: Summary of public-safe FCN evidence for the result reported in this paper.

Further public-safe quantitative observations recorded in the FCN entry are summarised below.

- H(BF16_value) = H(sign) + H(exponent) + H(mantissa) - MI(exp;mant)
- = 10.536 bits
- Exponent: 2.611 bits / 8 bits → 67% compressible
- Mantissa: 6.972 bits / 7 bits → 0.4% compressible
- Sign: 1.000 bits / 1 bit → 0% compressible
- 60+ codecs tested, ALL hit this ceiling

4.1 Public-safe restatement: Derived Formula

H(BF16_value) = H(sign) + H(exponent) + H(mantissa) - MI(exp;mant) $\approx$ 1.000 + 2.611 + 6.972 - 0.047 = 10.536 bits

r_max = 16 / H(value) = 16 / 10.536 $\approx$ 1.518x (practical) r_theoretical = 16 / 10.512 = 1.522x (Shannon limit)

Component breakdown: Exponent: 2.611 bits / 8 bits → 67% compressible Mantissa: 6.972 bits / 7 bits → 0.4% compressible Sign: 1.000 bits / 1 bit → 0% compressible

5 Discussion

The result is interpreted as a property of the trained-weight distribution rather than of any particular codec. The implication for compression is that any single-format, single-model lossless coder that targets only the value distribution is bounded by the same entropy term. Breaking the bound therefore requires moving out of that regime, by conditioning on additional context such as a paired base model, by changing the reconstruction target, or by changing the numeric format. The Forlais FCN evidence base records each of these regimes as separate entries with their own measurements and limitations.

6 Limitations

This paper has 4 limitations.

1. The reported value is an empirical constant verified across the tested architectures and is not a closed-form proof for every architecture.
2. The result holds for the tested numeric formats and may shift in adjacent formats with different mantissa widths.
3. The result is sensitive to training procedure; the verified architectures are mature, fully-trained networks.
4. Public reproducibility is limited by disclosure control; internal verification artefacts are maintained by Forlais Group but are not included in this paper.

7 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

8 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

9 Conclusion

The FCN evidence supports a verified, narrowly scoped statement: BF16 1.52x Entropy Ceiling is a stable feature of the tested trained-weight distributions and is not an artefact of any single codec. The result is reported here in its public-safe form and should be read alongside the limitations enumerated above.

References

[1] Elwyn R. Berlekamp. Algebraic coding theory. 1968.

[2] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006.

[3] David A. Huffman. A method for the construction of minimum-redundancy codes. *Proceedings of the IRE*, 40(9):1098-1101, 1952.

[4] IEEE. IEEE standard for floating-point arithmetic. IEEE Std 754-2019 (Revision of IEEE 754-2008), 2019.

[5] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023.

[6] Dhiraj Kalamkar, Dheevatsa Mudigere, Naveen Mellempudi, Dipankar Das, Kunal Banerjee, Sasikanth Avancha, Dharma Teja Vooturi, Nataraj Jammalamadaka, Jianyu Huang, Hector Yuen, Jiyan Yang, Jongsoo Park, Alexander Heinecke, Evangelos Georganas, Sudarshan Srinivasan, Abhisek Kundu, Misha Smelyanskiy, Bharat Kaul, and Pradeep Dubey. A study of BFLOAT16 for deep learning training, 2019.

[7] James L. Massey. Shift-register synthesis and BCH decoding. *IEEE Transactions on Information Theory*, 15(1):122-127, 1969.

[8] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations*, 2018.

[9] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners, 2019. OpenAI technical report.

[10] Jorma Rissanen and Glen G. Langdon. Arithmetic coding. *IBM Journal of Research and Development*, 23(2):149-162, 1979.

[11] Amir Said. Introduction to arithmetic coding: Theory and practice. Technical Report HPL-2004-76, Hewlett-Packard Laboratories, 2004.

[12] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379-423, 1948.

[13] Ian H. Witten, Radford M. Neal, and John G. Cleary. Arithmetic coding for data compression. *Communications of the ACM*, 30(6):520-540, 1987.

Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Expanded references, structured evidence summary, scope, and limitations.	Oliver Cooper
0.3	14 May 2026	Full evidence-grounded manuscript with public-safe abstraction, table summaries, and category-aware framing.	Oliver Cooper